

YERIN CHOE

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RESEARCH INTEREST

Advanced reductive water treatment, electrochemical catalyst, determination and degradation of Per- and Polyfluoroalkyl Substances (PFASs), sustainability, scalable process, photochemical degradation, tandem process, non-noble metal catalyst

EDUCATION

The University of Texas at Austin (UT Austin), Austin, TX Aug. 2025 – Present
PhD student in Civil Engineering

Seoul National University (SNU), Seoul, South Korea Mar. 2022 – Feb. 2024
Master of Science in Civil and Environmental Engineering
Thesis: Investigation on the applicability of UV/sulfite reductive treatment of PFASs in sorption regenerant

Seoul National University (SNU), Seoul, South Korea Mar. 2017 – Feb. 2022
Bachelor of Science in Chemistry (cum laude)
Thesis: Synthesis of less rigid ^{acri}PNP ligand to offer enhanced reactivity toward carbonyl conversion

PUBLICATION

1. Heejin Yang, Eunha Kim, Seong-Jik Park, Jaehyeong Park, **Yerin Choe**, Jong Kwon Choe, Chang-Gu Lee, Pedro J.J. Alvarez (2026) Hydroxide ions enhance a new reductive pathway for PFOA degradation in UV/sulfite processes, *Separation and Purification Technology* 406, 139085. <https://doi.org/10.1016/j.seppur.2026.139085>
2. Jaehee Kim, Taeyeon Kim, Heungjoo Park, Moon-Kyung Kim, Soyeon Eom, **Yerin Choe**, Jong Kwon Choe, Kyung-Duk Zoh (2024) Kinetics and proposed mechanisms of hexafluoropropylene oxide dimer acid (GenX) degradation via vacuum-UV (VUV) photolysis and VUV/sulfite processes, *Journal of Hazardous Materials* 463, 132864. <https://doi.org/10.1016/j.jhazmat.2023.132864>

RESEARCH EXPERIENCE

Werth Research Group (Prof. Charles Werth), UT Austin Aug. 2025 – Present
Graduate Research Assistant (Aug. 2025 – Present)

- Investigated electrochemical catalyst for reductive defluorination of PFAS
- Investigated electrocatalytic nitrate removal using parallel plate thin layer flow reactor

Laboratory of Separation and Reaction Engineering-Laboratory of Catalysis and Materials (Dr. Salomé Soares), University of Porto May 2026 – Jun. 2026
Exchange researcher

- Characterized and explored carbon nanotube (CNT)-supported metal catalysts for electrochemical degradation of PFAS

Sustainable Environmental Technology Lab (Prof. Jong Kwon Choe), SNU Mar. 2021 – Feb. 2025
Post-master Fellow (Mar. 2024 – Feb. 2025)
Research Assistant (Mar. 2022 – Feb. 2024)
Undergraduate Research Intern (Mar. 2021 – Feb. 2022)

- Conducted batch and column tests to investigate sorption removal of Per- and Polyfluoroalkyl Substances (PFASs) in industrial wastewater
- Examined inhibition mechanisms of sorption regenerants during UV/sulfite treatment of PFASs
- Compared field-implementation strategies to mitigate inhibition caused by regenerant components in UV/sulfite treatment of PFASs
- Enhanced nuclear magnetic resonance (NMR) data quality using principal component analysis (PCA)

Metabolism-Carbon Modeling Group (Prof. Jooyoung Park), SNU Mar. 2024 – Dec. 2024
Research Intern

- Conducted annual MFA of food-related nitrogen in Korea
- Incorporated ten processes for MFA including crop production, feed industry, human consumption, water treatment, etc.

Inorganic Molecular Conversion Lab (Prof. Yunho Lee), SNU Jun. 2020 – Feb. 2021
Undergraduate Research Intern

- Synthesized phenyl derivative of ^{acri}PNP ligand for the reduction of CO₂ into carbonyl compounds

Synthetic Organic Chemistry Lab (Prof. Hong Geun Lee), SNU Jun. 2019 – Feb. 2020
Undergraduate Research Intern

- Synthesized chiral ligand to induce enantioselective N-centered conjugate addition to activated alkene

PRESENTATION

[Oral]

- **Yerin Choe & Jong Kwon Choe** (2024, November 6-8). *The impact of organic solvents and inorganic salts on the treatment of per- and polyfluoroalkyl substances and strategies for mitigation* [In-person oral presentation]. 2024 Korean Society of Environmental Engineers Conference, Yeosu, South Korea.
- **Yerin Choe & Jong Kwon Choe** (2023, August 16-21). *Influence of regenerant conditions on degradation and defluorination efficiency of UV/sulfite reductive treatment of PFASs in sorption regenerant* [In-person oral presentation]. ACS Fall 2023, San Francisco, CA, United States.

[Poster]

- **Yerin Choe & Jong Kwon Choe** (2022, November 8-11). *Investigation on photoreductive degradation products and degradation mechanisms of perfluorooctane sulfonate (PFOS) using ¹⁹F NMR characterization* [In-person poster session]. 2022 Korean Society of Environmental Engineers Conference, Jeju, South Korea.

PROJECT

Adsorption-based Removal Technology and Detoxification of PFASs

Feb. 2024 – Nov. 2024

Samsung Electronics

- Predicted the adsorbent exchange cycle of a company's wastewater treatment facility based on laboratory experiments

Comprehensive Compilation and Utilization Framework Development for Carbon Neutral Technologies Information: Energy Sector

Aug. 2024

- Collected process flows and technical information in oil-refinery sector to develop reference energy system (RES)

Super Recalcitrant PFAS Treatment LAB

Mar. 2021 – Feb. 2024

National Research Foundation of Korea

- Investigated photoreductive degradation of PFASs in UV/sulfite process

Precise Analysis of PFCs in G-tower Wastewater

Jul. 2023 – Dec. 2023

Samsung Electronics

- Developed ¹⁹F NMR analysis method for detection and quantification of PFASs in wastewater

LEADERSHIP / EXTRACURRICULAR

Clean Water Science Network

Feb. 2026 – Jun. 2026

- Mentored two students from Latin America on scientific research methodologies, graduate school life, and career opportunities in the water sector

2026 STEM Girl Day, UT Austin

Feb. 2026

- Assisted visitors at the information tent as a individual volunteer

Volunteering Engineers & Scientists of SNU, SNU

Mar. 2021 – Feb. 2022

- Developed assistive device for improving mobility rights of visually impaired individuals

College of Natural Sciences Student Government, SNU

Sep. 2017 – Nov. 2019

Vice President (Oct. 2018 – Mar. 2019), Team leader (Apr. 2019 – Nov. 2019)

- Served as Vice President, directly elected by the undergraduate student body of the college

Tenspoon (Non Profit Organization), SNU

Sep. 2018 – Feb. 2019

- Volunteered weekly at the university cafeteria in exchange of meal tickets for students in financial need

Summer Science Volunteering Camp, SNU

Aug. 2017, Aug. 2018

Organizing committee

- Participated in planning and volunteer works for science education and mentoring programs aimed at rural youth

SNU Girls' Dance Crew (GoAhead), SNU

Mar. 2017 – Dec. 2022

Vice President (Mar. 2018 – Feb. 2019)

- Directed five performance showcases involving 30 to 50 members as Vice President

HONORS AND AWARDS

Water Environment Association of Texas (WEAT) - Student Design Competition Apr. 2026
WEAT

Awarded 3rd place for designing wastewater treatment plant in El Paso, Texas

College Recruitment Fellowship Fall 2025 - Present
UT Austin

Conference Paper Award Nov. 2025
Korean Society of Environmental Engineers
For oral presentation at the 2024 KSEE Conference

Alumni Association Scholarship Fall 2023, Spring 2023, Fall 2022
Gwanak Corporation, SNU
Awarded to 3 graduate students in the department for studies in water treatment

BrainKorea21 Four Scholarship Spring 2022
National Research Foundation of Korea

Merit-Based Scholarship Spring 2018, Fall 2020
SNU
Awarded 50% tuition in Spring 2018, 30% tuition in Fall 2020

Alumni Association Scholarship Spring 2019
Gwanak Corporation, SNU
Awarded to 10 undergraduate students in the university for remarkable dedication to student government

13th Creative Design Competition for Underserved Communities 2021
Sharing and Technologies Incorporated
Awarded Bronze Prize for developing an assistive device for visually impaired individuals

SKILLS

TEM, LC/MS, LC/MS/MS, IC, GC/MS, ICP/MS, ICP/OES, NMR (1D & 2D, experience in 1H, 13C, 19F, 31P), TOC analysis, SPE, organic synthesis, openLCA, STAN, GPS-X